UNIVERSIDAD DE LAS FUERZAS ARMADAS ESPE

Department of Computer Science

Software Engineering Degree

Advanced Web Development

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Project: Fábula Dental Management System

Problem:

The following problems arise within a children's dental clinic, called The Fábula Dental.

A system is needed to manage the inventory of medical supplies to prevent waste.

Additionally, the system needs to manage patient cash flow within the clinic.

The system must prevent the waste of expired medications (unnecessary purchases) and automate the control of patient credit balances, eliminating reliance on text messages or easily lost manual records.

Overview

In the context of managing the administration of a children's dental practice like Fábula Dental, it is essential to have tools that allow for the efficient organization and control of internal resources and processes. In particular, the proper management of medical supply inventory and the control of patient payments are key aspects for the practice's smooth operation.

The proposed system aims to offer a web-based solution that centralizes and automates these processes. On the one hand, it will allow for the management of medication and supply inventory, facilitating the registration, consultation, and updating of stock levels, in order to avoid unnecessary purchases and reduce product waste. On the other hand, the system will allow

for detailed tracking of patient payments, including partial payments, procedures performed, and available balances. This will allow for an orderly tracking of each patient's payment transactions, avoiding confusion or errors.

Background

To ensure the system's effectiveness, it is necessary to define clear processes for handling dental supplies and patient payments within the clinic.

- **Medical Supplies and Inventory**

  Medical supplies and medications are valuable assets for the clinic. Unused or expired medications represent a direct loss of resources. To prevent this and maintain a clear history of what is in storage, the system will assign unique names or codes to every item, replacing easily lost manual records with an organized digital catalog.

- **Patient Payments and Credits**

  Patients usually make an initial payment when starting a treatment. For accurate administration, the clinic must link these advance payments to the specific procedures performed later. Until the dental service is completed, the patient's initial deposit acts as an available credit balance in their favor.

Patient and Inventory Tracking

Effective clinic management requires a precise record of all product usage and financial movements. The system will automate this control by creating a complete history that includes the following data:

- **Item Name:** A clear identifier for each medication or supply.
- **Quantity:** The number of items purchased or consumed.
- **Dates:** Purchase dates and expiration dates to monitor product life cycles.
- **Costs and Prices:** This includes the price of buying dental materials and the cost of the services provided to the patient.
- **Patient ID:** A unique identifier linking deposits and treatments to the correct patient.

System Metrics

The system will show the real status of the clinic's inventory and patient accounts using these basic measures:

- **Total Balance:** The sum of all deposits made by a patient minus the total cost of their completed procedures.
- **Inventory Alerts:** Automated notifications based on the comparison between items bought and used, showing current stock to avoid buying unnecessary supplies and eliminating the reliance on informal text messages.

This system lets the staff quickly see which patients have credit and which medicines they need to buy soon. This helps Fábula Dental run better and avoid mistakes.